# Kongarasan Sathiya Moorthy

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## Profile Summary

Master's student in Computer Science specializing in AI Engineering, Generative AI, and machine learning experimentation. Hands-on experience in LLM pipelines, RAG systems, and model evaluation workflows, with a strong focus on benchmarking, prompt-based generation, and structured experimentation. Skilled in Python, transformer models, and analyzing AI model performance for practical applications.

## Education

**Master of Science in Computer Science**
University of Passau, Germany 2024 – Present

- **Relevant Coursework:** Deep Learning, Computational Linguistics, Computer Vision

**Bachelor of Technology in Computer Science and Engineering**
SRM Institute of Science and Technology, India 2020 – 2024

## Experience

**Software Engineering Intern** Mar 2023 – Feb 2024
*Triotek Nanocare Solutions, India*

- Built a single-page web application using HTML, Tailwind CSS, and JavaScript following user research and high-fidelity UI design.
- Deployed packages and configured an Nginx reverse proxy for the backend API.
- Contributed to a UI redesign and React development for the company's mobile app.

## Projects

**RAG-based PDF Chatbot (Generative AI + Evaluation Pipeline)**
*Python, LangChain, ChromaDB, Streamlit, LLaMA*

- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline using embeddings and semantic search
- Built document ingestion, chunking, and vector indexing pipelines for scalable retrieval workflows
- Conducted structured and reproducible evaluation of model outputs using defined quality metrics
- Experimented with prompt variations and retrieval strategies to optimize generation quality

**AI Resume Analyzer (LLM + Model Evaluation System)**
*Python, FastAPI, Streamlit, Gemini API, Pandas*

- Developed an LLM-based system for structured text evaluation and automated scoring workflows
- Built end-to-end NLP pipelines for data extraction, preprocessing, and model input preparation
- Designed reproducible experimentation workflows to evaluate prompt strategies and model behavior
- Generated synthetic datasets to evaluate robustness and consistency of model outputs

**AI Workflow Automation (LLM Systems + Model Benchmarking)**
*n8n, REST APIs, GPT-based systems*

- Designed automated AI pipelines integrating multiple LLM APIs for decision-making workflows
- Built agent-like systems for task automation and structured data processing
- Evaluated and benchmarked multiple language models using comparative performance metrics and analysis
- Compared model outputs to identify optimal configurations across use cases

## Skills

**Programming Languages:** Python, SQL
**Machine Learning & NLP:** ML, NLP, Transformers (BERT), Embeddings, RAG, Fine-tuning, Prompt Engineering
**AI Research & Evaluation:** Model Evaluation, Benchmarking, Prompt Testing, Experimental Design, Synthetic Data Generation
**AI Engineering:** LangChain, LLM APIs, Pipeline Design, Model Integration, PyTorch
**Data & Algorithms:** Pandas, NumPy, EDA, Algorithmic Problem Solving
**Tools & Frameworks:** FastAPI, Scikit-learn, Hugging Face (basic), REST APIs, Git
**Databases:** PostgreSQL, ChromaDB (Vector DB)
**Cloud Platforms:** Google Cloud Platform, Microsoft Azure
**Computer Vision:** Image Processing, Feature Extraction, Texture Analysis

## Certifications

- AI for Everyone, DeepLearning.AI / Oracle Academy (Mar 2023)
- Cloud Computing — Upskillz (Nov 2022)

## Languages

English – Fluent | German – A2 (currently pursuing B1)